Evaluating the influence of magmatic sulphides on chalcophile element enrichments in the Bradbury assemblage, Gale Crater, Mars, using APXS measurements

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Introduction: Among the rocks encountered by the Mars rover, Curiosity, are igneous rocks and (nearly) isochemically-derived sediments that appear to reflect two magmatic reservoirs: Jake-M class rocks characterized by elevated Na_2O, K_2O, Al_2O_3 and SiO_2, and the Bathurst Inlet class characterized by high MgO, FeO_{total}, MnO, and Zn and low SiO_2 [1]. Sedimentary processes, chemical weathering reactions, and hydrothermal alteration of source rocks appear to have affected little mineralogical change in many Bradbury group rocks analyzed by the Curiosity rover [2]. This may point to a role of magmatic sulphide mineralization in the formation of the enrichments in Zn and other chalcophile elements in the Bradbury Group [3]. Primary magmatic sulphides would have been subjected to postemplacement hydrothermal redistribution of mobile elements. Weathering and transport of magmatic and hydrothermal sulphides could then lead to the dispersal of chalcophile element rich material into the Bradbury assemblage sediments.

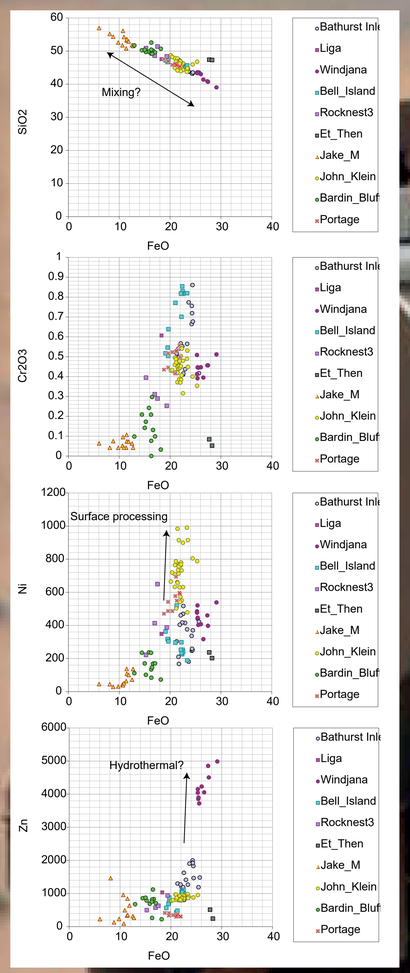
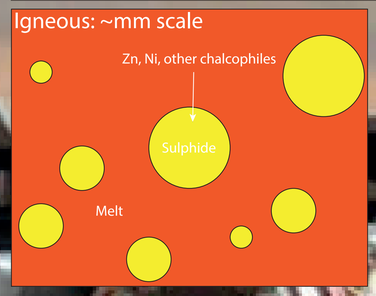


Figure 1: Variation diagrams of SiO_2, Cr_2O_3, Ni and Zn vs.FeO_T.

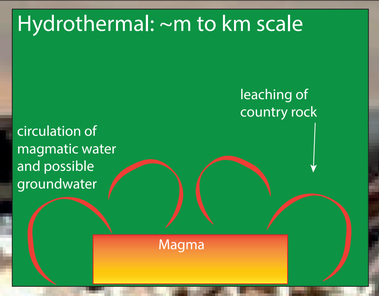
Influences on chalcophile element distributions in Gale crater

Igneous: ~mm scale



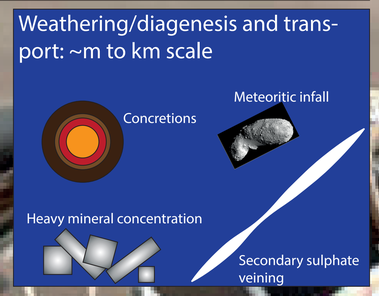
Igneous sulphides precipitate directly from melt, concentrating chalcophile elements. Pyrrhotite and pyrite have been detected by CheMin, as well as akaganite (a weathering product of Fe-sulphides). Sulphides may remain disseminated, or may accumulate (e.g., by density segregation) into more massive bodies. Assimilation of crustal S may be important in the formation of such sulphides.

Hydrothermal: ~m to km scale



Hydrothermal solutions (magmatic or near-surface fluids) react with and metasomatize crustal rocks, leaching chalcophiles from country rock and precipitating new sulphide minerals. Both endogenic and exogenic (impact) heat sources may be involved. Differences in mobility may cause decoupling of some elements, e.g., Zn and Mn, from parental igneous trends.

Weathering/diagenesis and transport: ~m to km scale



Veins, concretions, and secondary minerals form via hydrolysis of sulphides. Elements that are concentrated in heavy, resistant minerals (e.g., Cr in chromite, Ti in ilmenite) may be concentrated by sedimetary processes. Meteorite infall is expected to add siderophile and some chalcophile elements, notably Ni.

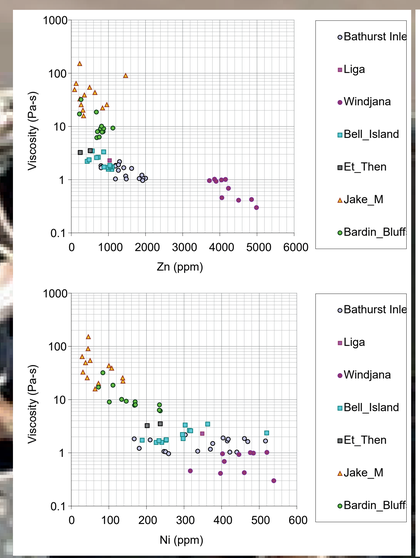


Figure 2: Upper bound case for Sulphide Concentration at Sulphide Saturation (SCSS), T=1350 °C, P=0.1 GPa, all Fe as FeO, and measured composition first normalized to 100% then renormalized without SO_3 and Cl. Formulation of Ding et al. [5]:

$\ln(SCSS) = 14.25 - 3948.51/T - 0.06P - 2.81[SiO_2]_{melt} - 7.33[Al_2O_3]_{melt} + 0.97\ln([FeO]_{melt})$.

Calculated SCSS values correlate with chalcophile elements Ni and Zn. Soils and sediments show decoupling of Ni from Zn, possibly due to different mobilities of these elements in hydrothermal or weathering processes, or the concentration of Ni in mudstones from meteoritic infall and Ni accumulation in Mg-sulphates.

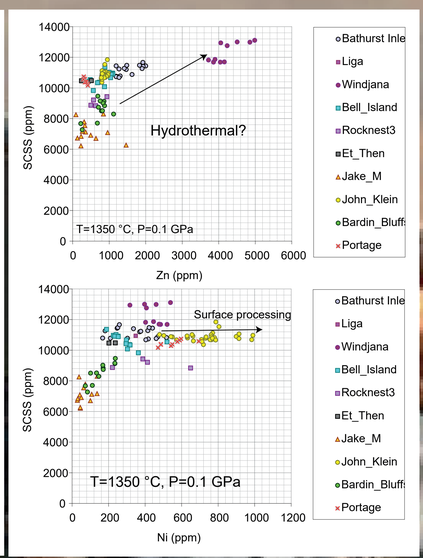


Figure 3: Calculated melt viscosity for APXS analyses thought to represent plausible melt compositions (soils and John Klein mudstones excluded) using the formulation of Shaw et al. [14]. The inverse relationship of viscosity with Ni/TiO2 ratio is consistent with concentration of Ni in lower-viscosity melts. Viscosity is strongly dependent on SiO_2 and FeO concentrations and the anticorrelation of viscosity and [Ni] in part reflects this, but this is also consistent with enhanced sulphide segregation in the low viscosity low SiO_2, high FeO melts.

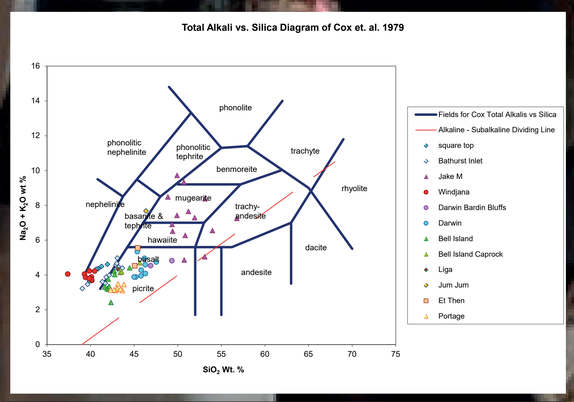


Figure 4: Total alkalis vs. silica diagram for a selection of Bradbury Assemblage rocks analyzed by APXS. The alarge range in total alkali concentration in Jake M class rocks is conspicuous, and may be due to variations in the source region or differences in the fractionation pathways during emplacement.

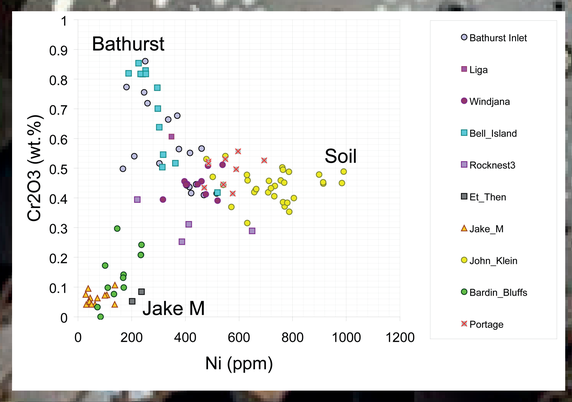


Figure 5: Variation diagram with Ni and Cr_2O_3 concentrations, showing that compositional variations in many of the Bradbury group rocks could be explicable by mixing between Jake M, Bathurst, and average local soil components.

Methods: To assess the potential role of magmatic sulphide mineralization in the Bradbury assemblage, we have used whole-rock chemistry measurements from the Alpha Particle X-ray Spectrometer (APXS) instrument [4]. Estimated Sulphide Concentrations at Sulphide Saturation (SCSS) values were calculated according to the formulations of Ding et al. [5] and Righter et al. [6] from the APXS-measured compositions. Because the empirical equations of Ding et al. are calibrated for FeO-rich compositions, we have used this formulation in this work. Because there are no measurements of iron valence for the analyzed rocks, we take assume that all Fe occurs as FeO. Further, the APXS analyses are initally normalized to 100%, and then recalculated without SO_3 and Cl following common practice [e.g., 7]. Therefore, the SCSS values calculated here represent upper bounds. Viscosities were calculated for anhydrous melts of the APXS-measured compositions (normalized to 100%, then renormalized with SO_3 and Cl set to 0 with all Fe as Fe^{2+}), and therefore represent upper bounds. The addition of water and other volatiles would be expected to decrease melt viscosity. The assumed temperature was 1270°C, changing the temperature will affect the quantitative but not the qualitative character of the relationships.

Discussion:

- Estimated sulphide concentration at sulphide saturation for Gale crater igneous rocks and isochemically-altered sediments, including Jake_M and Bathurst, are up to ~13500 ppm (Fig. 1), depending on the choice of parameters used. **Qualitatively, these SCSS calculations show that Jake_M and Bathurst Inlet melts would have been efficient carriers of sulphur and chalcophile elements**.
- Upon sulphide super-saturation, low viscosity will enhance the segregation of immiscible sulphide melts. **Calculated viscosity for measured igneous rock compositions shows an inverse relationship between melt viscosity and concentrations of Ni, Cr, and Zn**.
- The relationship with Cr may indicate a segregation of dense oxide (Cr-spinel) or be coincidental due to the mafic to ultramafic character of the Bathurst-type melts.
- **Assimilation of crustal sulphur may have caused the primary magmas to reach sulphide saturation** [8]. Superheated impact melts might be able to erode and assimilate 'country rock' and may be ideally suited to form 'magmatic' sulphide deposits. Impact melts are also likely to be highdly inviscid, which can enhance the segregation of immiscible sulphides.

Conclusions: The initial formation of magmatic sulphides during the emplacement of Martian melts (of endogenic and impact origin) can provide initial enrichment in chalcophile elements, which can later be mobilized and redistributed by hydrothermal, weathering and sedimentary processes.

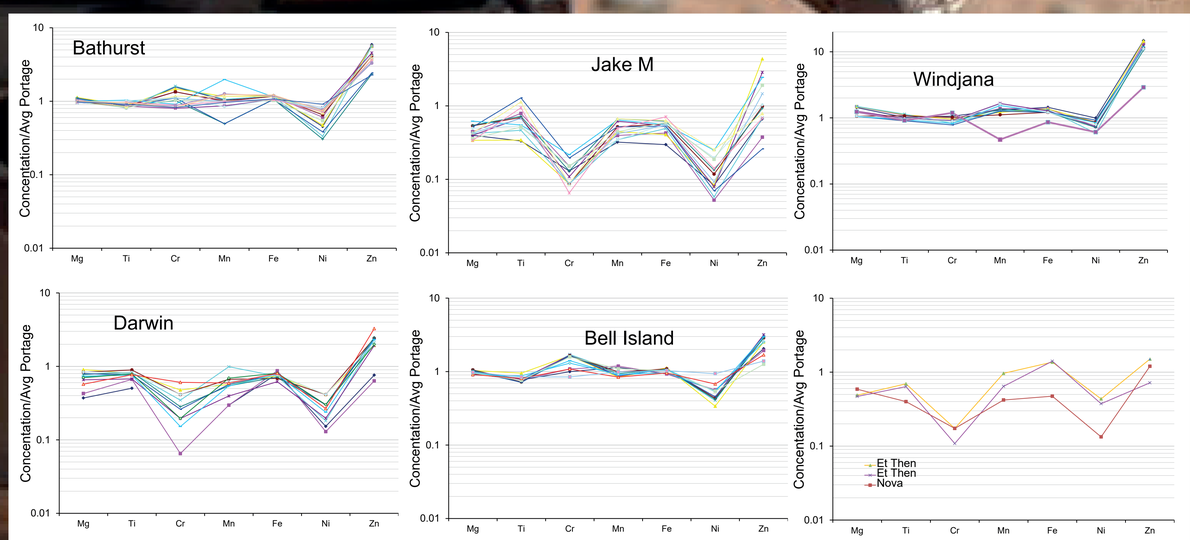


Figure 6: Multielement plots of chalcophile elements normalized to average Portage soil. Most rock analyses show an enrichment in Zn and depletion in Ni, most notably the Jake M and Darwin classes. Bathurst, Windjana and Bell Island are comparatively flat, possibly indicating that these rocks are more representative of the average input to Portage soils. Jake M and Darwin analyses are generally depleted in Cr and Ni relative to Portage soil, this may reflect lower abundances of the igneous carriers of those elements (Ni in olivine, Cr in chromite/chrome spinel and possibly Cr-bearing clinopyroxene). The Et Then, Liga, and Nova analyses are somewhat similar to Jake M and Darwin, although Liga is Cr-rich. Mn and Cr show considerable variability within sample classes. Overall, the chalcophile elements Ni and Cr appear to reflect primary (silicate) igneous processes, while Zn may have been enriched by a combination of magmatic and hydrothermal processes. Zinc may be present as the ZnS molecule, either as a solid solution within Fe-sulphides (e.g., pyrrhotite, pyrite) or as sphalerite. Further investigation of APXS spectra may reveal other chalcophile elements not currently reported (e.g., Cu, Co, Ge) that may further elucidate the role of sulphide minerals in the geochemistry of Gale Crater rocks.

References: [1] Schmidt et al., 2014 J. Geophys. Res. Planets, 119(1). [2] Treiman et al., 2016 J. Geophys. Res. Planets, accepted. [3] Berger et al., 2015 LPSC XLVI Abs. #1564. [4] Gellert et al. 2015 Elements, 11 (1), 39-44. [5] Ding et al., 2014 GCA 131 (2014) 227–246. [6] Righer et al., 2009 EPSL 288 235–243. [7] Ming et al. 2008 J. Geophys. Res. Planets, 13, E12S39. [8] Baumgartner et al. 2015 Ore Geol. Reviews 65 400–412. [9] Agee et al. 2013 Science 339, 780–785. [10] MacAdam et al. 2014 J. Geophys. Res., Planets 119, 373-393. [11] [12] Stolper et al., 2013 Science 341(6153). [13] Schmidt et al. 2016 this meeting [14] Shaw 1972 Am. J. Sci. 272, 870-893.

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Alternate title: Magmatic Sulphides Caused Life-sustaining Chalcophile Element Enrichments in the Bradbury Assemblage, Gale Crater, Mars, Using APXS Measurements: Implications for habitability on Europa?